

A mixed integer linear programming model for quarantine-based home healthcare scheduling under uncertainty

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ABSTRACT

Home healthcare companies (HHC) have emerged as vital alternatives to traditional hospitals, particularly in meeting the healthcare needs of individuals within the comfort of their homes. The COVID-19 pandemic has amplified the significance of HHC services, offering a crucial alternative for patients and the elderly to follow quarantine protocols while receiving essential healthcare at home. Consequently, HHC companies must align their planning strategies with the World Health Organization (WHO) health guidelines. This research introduces a Mixed Integer Linear Programming (MILP) model tailored for home healthcare services during COVID-19, aiming to ensure strict adherence to quarantine protocols while enhancing service efficiency and quality. The proposed vehicle routing problem with pickup/delivery and time window formulation incorporates critical elements such as patient and caregiver classification, work and break regulations adherence, workload balancing, and multi-depot capabilities. The model addresses uncertain demand and service times through a stochastic programming approach to enhance practicality. K-means clustering is applied to streamline scenarios, with a sensitivity analysis determining the optimal number of clusters. Additionally, measures intrinsic to stochastic programming, such as the Expected Value of Perfect Information (EVPI) and Value of Stochastic Solution (VSS), are computed for comprehensive analysis.

1. Introduction

A healthcare management system, crucial for addressing health service needs, faces challenges posed by infectious diseases, including the global impact of the COVID-19 pandemic [1,2]. A wide range of comprehensive strategies, including public health measures, testing and tracing, vaccination programs, and targeted interventions, have been implemented through collaborative efforts to mitigate the impact of COVID-19 and limit its transmission [3].

The World Health Organization (WHO) advocates preventive measures, and a "bring-service-near-your-home" approach [4] is suggested to provide essential medical support while ensuring compliance with isolation guidelines [5]. Simultaneously, home health care (HHC) emerges as a viable solution to contemporary healthcare challenges, offering cost-effective and convenient alternatives to hospital-based services [6]. The optimization of the Home Health Care Routing and Scheduling Problem (HHCRSP) is pivotal for HHC companies to enhance overall efficiency amid challenges like high operational costs and limited resources [7].

The COVID-19 outbreak has disrupted businesses, especially those reliant on customer mobility in densely populated areas [8]. Home healthcare companies, facing challenges from the pandemic, must adapt their planning methods due to uncertainties in the number of applicants and service times. Practical scenarios, influenced by the unpredictable nature of the corona disease, introduce uncertainty until caregivers reach patients' homes. This uncertainty, coupled with potential delays, poses risks to patient satisfaction and the reputation of HHC companies. Additionally, HHC patients often require home delivery services (HDSs), crucial for providing seamless healthcare experiences, especially during the pandemic [9]. Also, HHC companies engage in diverse logistics activities, encompassing the delivery of medications and medical devices from pharmacies to patients and the transportation of biological samples from patients' homes to laboratories [10]. Addressing uncertainties in demands and service times becomes essential for effective planning and execution in HHC. Particularly amidst the COVID-19 period, it's crucial to take into account the unpredictable fluctuations in demand within patients' households and the connection between demand and service durations. Due to the nature of COVID-19, the symptoms of the disease

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can be revealed suddenly, especially for people living in the same household. As a result, when caregivers arrive at a patient's home, they may be faced with more than one applicant for receiving medical care. This sudden increase in the number of demands and service time required at the patient's home poses a significant challenge that cannot be adequately addressed by traditional deterministic models. The proposed stochastic programming approach, which incorporates the uncertainty of demand and service times, is better equipped to handle these dynamic and unpredictable situations, ensuring a more effective and efficient healthcare delivery system during the pandemic.

Motivation: In the context of the COVID-19 pandemic, where compliance with quarantine and social distancing measures is paramount and considering the inherent uncertainties in demands and service times, the integration of these factors into a comprehensive mathematical model for nurse assignment, scheduling, and routing becomes imperative. This integration aims at maximizing efficiency, minimizing costs, and adapting to the unique challenges posed by the pandemic. This paper seeks to address several pivotal research questions pertaining to the impact of the COVID-19 outbreak on healthcare delivery within the realm of home healthcare. These questions include.

1. How can HHC companies strategically position themselves to contribute to society's efforts in mitigating the spread of the disease during critical situations?
2. In what ways can HHC companies enhance their effectiveness in curbing the spread of disease by modifying their planning methods?
3. What criteria should HHC companies prioritize to enhance the satisfaction of both caregivers and patients during critical situations?
4. How can uncertain parameters be incorporated into the planning model of HHC to ensure adaptability to real-world conditions?
5. What is the impact of simultaneous consideration of a stochastic parameter (demand) with a discrete distribution and another stochastic parameter (time service) with a continuous distribution on a decision variable?
6. How can a balanced trade-off be achieved between quantitative factors and qualitative criteria in decision-making processes?

Research gap: To the best of the authors' knowledge, there is no paper in the literature considering the WHO protocols to prevent the spread of Corona virus in HHC daily programming practically. So, this is the first model deeming quarantine and social distancing measures and the bring-service-near-your-home approach as a mathematical model under uncertainties.

Contribution: This research introduces an innovative mathematical model aimed at addressing the complexities inherent in the home healthcare routing and scheduling problem. The model incorporates critical aspects to enhance the overall efficiency and effectiveness of home healthcare services, including.

- Integrating WHO protocols into the mathematical framework as constraints, with the aim of elevating the health status of both caregivers and patients.
- Prioritizing caregivers' satisfaction and well-being by minimizing their presence in densely populated areas such as laboratories and pharmacies.
- Ensuring fairness among caregivers through the consideration of work regulations and workload balance.
- Optimizing the delivery of patients' healthcare requirements through the simultaneous utilization of vehicle services and care services. This approach not only reduces operational costs but also enables caregivers to supervise all vehicle services.
- Addressing the real-world nature of the problem by treating the number of demands in patients' homes and service times as stochastic parameters. This acknowledges the inherent uncertainty in these factors and enhances the model's applicability to practical scenarios.

The remainder of this paper is structured as follows: Section 2 provides an extensive review of pertinent studies in the field. Section 3 delineates the problem at hand and introduces the mathematical model. Section 4 elucidates the approach taken to derive solutions. Section 5 delves into the presentation of experimental results and a sensitivity analysis. Section 6 imparts managerial insights gleaned from the research findings. Finally, Section 7 brings the paper to a conclusion, encapsulating key takeaways and proposing avenues for future research.

2. Literature review

The Home Healthcare Routing and Scheduling Problem (HHCRSP) has garnered widespread attention from scholars worldwide over the past two decades [11]. Originating in 1974, Fernandez et al. introduced this intricate vehicle routing problem, imbued with additional constraints specific to home health care [12]. Notably, HHC routing and scheduling problems exhibit nuanced variations, encompassing diverse constraints like planning periods, depot numbers, time window restrictions, nurses' skills, work and rest rules, and multiple objective functions. In 1998, Cheng and Rich delved into considerations of time windows, multiple depots, and two distinct groups of part-time and full-time nurses [13]. Fikar and Hirsch contributed by exploring single- and multi-period home health care issues within the HHCRSP context [7]. Liu et al. formulated a multi-objective home healthcare routing and scheduling problem that tactically assigns caregivers to patients at home, factoring in practical constraints and optimizing three objectives: minimizing operating costs and maximizing patient and caregiver satisfaction [14]. Addressing downgrading concerns, Khodabandeh et al. proposed a mathematical model to minimize the disparity between nurses' actual and potential skills, employing the epsilon constraint [15]. Castillo et al. proposed a new mathematical model to assess the adequacy of providing home healthcare (HHC) systems in Spanish rural areas to avoid unnecessary trips during the COVID-19 era, and applied a Multi-start Biased-Randomised algorithm to solve this optimization problem [16].

2.1. HHC with workload balancing

Efficient workload balancing constitutes a pivotal element in the optimization models of the HHCRSP. The rationale behind workload equilibrium lies in its capacity to foster fairness among employees, thereby enhancing overall job satisfaction [17]. This employee satisfaction, in turn, assumes a crucial role in elevating service quality during interactions with customers. Furthermore, a balanced workload contributes to the dependability of pathway programs, as it mitigates service delivery failures and curtails overtime requirements [18]. Notably, Matl et al. (2019) underscored that there exists no universal scaling method for workload balancing [19]. The pertinent literature on this subject showcases diverse criteria for measuring workload and varying functions to define workload balance. For instance, Decerle et al. (2019) delineate workload as the working time of a caregiver, optimizing to minimize the maximum difference in working time among caregivers [20]. In a separate endeavor, Kandakoglu et al. (2020) developed the Home Dialysis Scheduler System (HDSS), a decision support system that considers factors such as nurses' workload balance, overtime work, the necessity for mealtime breaks, restrictions and preferences linked to visit times, and the diverse types of services provided to patients [21]. In a recent contribution, Liu et al. (2021) presented a multi-period, multi-objective home health care routing and scheduling problem, wherein appointments, lunch breaks, continuity of care, and workload balance are integrated as constraints [6].

2.2. HHC with synchronization

In the investigation of HHC operations, Bredstrom and Ronnqvist (2008) underscored the significance and intricacy associated with time

precedence and simultaneity issues [22]. Their focus centered on scenarios where multiple nurses are required to attend to a patient in a specific sequence, prompting the utilization of both meta-heuristic and exact methods to address this combined problem. Expanding on this theme, Bachouch et al. (2011) formulated an exact Mixed Integer Programming (MIP) approach, specifically addressing shared visits and interruptions [23]. Rasmussen et al. (2012) further delineated five distinct types of temporal precedence and concurrency issues within HHC operations [24]. Liu et al. (2018) contributed to the field by introducing a specialized vehicle routing problem, extending the classical problem by incorporating time window and synchronized-services constraints [25]. Wirnitzer et al. (2016) proposed five MIP formulations aimed at assessing various aspects of continuity of care in a mid-term home care cataloging problem [26]. In 2019, Decerle et al. presented an algorithm that amalgamates memetic methods and ant colony optimization, considering concurrency, workload balance, and time windows [20]. Pereira et al. (2020) delved into a novel realm by introducing a multi-period workforce scheduling and routing problem with dependent tasks, offering both a mixed integer programming model and heuristic algorithms for resolution [27].

2.3. HHC with pickup/delivery

The HHCRSP has evolved to encompass extended planning horizons, spanning weekly or monthly durations, to align with practical requirements. In real-world scenarios, the recipients of HHC services, including patients and isolated elderly residents, not only necessitate healthcare assistance but also demand home delivery services (HDSs) for medications, meals, and health test reports [9]. In the domain of chemotherapy operations at home, Salma et al. (2009) directed their attention to the intricate anti-cancer drug supply chain [28]. Liu et al. (2012) ventured into the development of mathematical models pertinent to home health care companies engaged in the seamless execution of pickup and delivery tasks to fulfill the diverse demands of patients [8]. Nasir et al. [9] contributed by formulating an innovative mathematical model that accommodates multiple synchronized home delivery services, pickup-delivery visit-based precedence constraints, and schedules for the delivery vehicles based on multiple routes. Abbasi et al. [29] proposed a new integrated location-allocation-routing model related to the home healthcare supply chain (HHCS) to make these decisions, and employed LP metric and epsilon constraint methods to solve this model. Mishra et al. [30] developed a new multi-objective model for the relief items scheduling problem, aiming to minimize the distribution time and operational cost under a heterogeneous vehicle scenario.

2.4. Studies under conditions of uncertainty

In home health care research, conventional methods assume precise data, predetermined without uncertainty. Liu et al. [6] addressed a multi-faceted home health care routing problem, while Khodabandeh et al. [31] created a model allowing flexibility in nurses' routes. Li et al. [32] focused on minimizing costs and waiting times while maximizing patient satisfaction. However, real-world conditions introduce randomness. Researchers like Lanzarone and Matta [33], Yuan et al. [34], Issabakhsh et al. [35], Demirbilek et al. [36], Carello et al. [37], and Liu et al. [38] addressed uncertainties using robust optimization methods and innovative modeling techniques. Shi et al. [39] and Yang et al. [40] incorporated fuzzy demand and probabilistic constraints. Demirbilek et al. [41] proposed a novel decision-making method, Yuan et al. [42] addressed daily caregiver scheduling with uncertainties, and Shahnejat-Bushehri et al. [43] developed a robust optimization model considering uncertain service times and travel times, temporal dependence, and synchronization constraints.

Nasiri et al. [44] investigated the uncertain demand in a green location-inventory-routing optimization model with simultaneous

pickup and delivery under disruption risks. Meanwhile, Ghabouh et al. [45] conducted a systematic literature review that found demand planning issues under uncertainty have received less attention from researchers in the home healthcare (HHC) problems domain. In recent studies, some researchers have utilized state-of-the-art methods, such as artificial intelligence (AI), the Internet of Things (IoT), machine learning (ML), and deep learning, to enhance healthcare systems [46,47]. These techniques have also been applied to various aspects of HHC, with AI and Operations Research (OR) proving to be effective in improving healthcare network operations [48]. For instance, Torabzade et al. [49] developed a new intelligent algorithm to assess the performance of an HHC center considering trust indicators and examined the effect of these indicators on patient satisfaction. Additionally, Nordfjærn et al. [50] investigated the impact of home time, work pressure, and fatigue on caregiver attention, applying regression methods to suggest effective measures to decrease caregiver job demands and maintain their health and safety at work.

This study proposes a distinct HHCRSP model to help reducing the crisis of quarantine conditions during a contagious illness with three key innovations. Firstly, it integrates quarantine and social distancing measures, designating multiple depots and categorizing nurses and patients to prevent COVID-19 spread. Secondly, it introduces stochastic parameters for demands and service times, a less-explored aspect in existing studies. Thirdly, the model emphasizes synchronization constraints for both HHC and home delivery services (HDSs), workload balancing, and work regulations, crucial for caregiver satisfaction, particularly during pandemic conditions. Additional features include constraints on lunch breaks, floating working start times, and hard time windows, making the proposed model stand out in addressing crisis situations such as the COVID-19 pandemic. Table 1 compares the most related papers in the literature on Home Health Care context.

3. Problem description and model formulation

3.1. Problem description

Considering uncertainties is essential in solving real-world problems, as observed by numerous researchers. HHC is a practical challenge in the health sector, and accounting for uncertainties in this context aids decision-makers in enhancing planning. Classical deterministic models of HHCRSP assume known variables, such as the number of patient requests and service times, beforehand. However, in reality, especially during the COVID-19 era, the unpredictable nature of demand in patients' homes and the correlation between demand and service times make deterministic models less reflective of actual scenarios.

The challenges posed by the pandemic, such as avoiding gatherings and protecting healthcare workers, as well as the need for a bring service near your home approach, require strategic planning. Particularly amidst the COVID-19 pandemic, it's crucial to take into account the volatility of demand in patients' households and the relationship between demand and service durations. The proposed model addresses these challenges by introducing a two-pronged approach: Home Healthcare (HHC) services administered by caregivers classified into corona and non-corona categories, and Home Delivery (HDSs) services facilitated by three types of vehicles to handle various tasks, including sample collection, medication delivery, and housekeeping services. This comprehensive solution helps minimize the risk of disease transmission while ensuring patients and caregivers can access the necessary healthcare and support services without having to venture into high-risk environments.

This study introduces randomness in the number of requests and service times, acknowledging the statistical distribution of demand. The HHC company's primary objective is to facilitate patient access to healthcare while minimizing costs and maximizing service quality. The challenges posed by the pandemic, such as avoiding gatherings and protecting healthcare workers, require strategic planning.

Table 1
The most related literature in the context of Home Health Care [51; 52; 53; 54].

Reference	No. of objectives		No. of depots		No. of periods		Assumptions of the model												Objective function						deterministic	Uncertainty								
	single	multi	single	multi	Single	multi	Time window	Delivery Time	Service consistency	Working time	Skill requirements	Caregiver consistency	Lunchbreak	synchronization	Task dependency	Pick up/delivery	Work regulation	Workload balancing	Travel balancing	Pickup/Delivery	Travel time/distance	Travel cost	Idle cost	Number of nurses/services		Nurse satisfaction	Patient satisfaction	demand	Travel time	Service time	stochastic	robust	fuzzy	
[1]	●			●	●		●		●		●		●	●				●					●			●	●	●						
[4]		●		●		●	●				●						●						●							●		●	●	
[5]	●			●	●		●				●												●							●	●		●	
[6]	●		●		●																		●						●	●	●			
[9]		●	●			●		●	●	●		●											●						●	●				
[14]		●	●		●		●				●												●					●						
[25]	●		●			●	●							●	●							●						●						
[26]	●		●		●		●				●			●	●	●	●						●		●			●						
[29]	●		●		●		●				●						●						●				●	●						
[38]	●		●			●	●		●		●			●	●	●							●		●					●	●		●	
[43]	●			●	●		●					●									●		●					●						
[44]	●		●		●		●															●							●		●	●		
[45]		●		●		●		●											●			●							●	●				●
This study	●			●	●	●				●	●		●	●	●		●			●			●	●					●		●	●		

The proposed model, as depicted in Fig. 1, addresses the challenges posed by the pandemic by introducing two distinct service types: Home Healthcare (HHC) services and Home Delivery Vehicle (HDV) services. The HHC services are administered by caregivers, who are classified into corona and non-corona categories, and each patient is assigned to a caregiver in the same category to prevent virus transmission. The caregivers start and end their routes from their homes, ensuring social distancing and minimizing the risk of disease transmission. The HDV services are facilitated by three heterogeneous vehicles, each with specific tasks: type one vehicles collect biological samples from patients' homes and deliver them to a laboratory, type two vehicles deliver medications and medical equipment, and type three vehicles offer housekeeping services, including meal delivery and laundry services. These HDV services help patients and caregivers avoid high-risk places such as laboratories and drugstores, further reducing the risk of disease transmission.

Given the interdependencies between HHC and HDSs services, the model emphasizes the necessity for coordinated scheduling and routing of both caregivers and vehicles, ensuring their simultaneous presence at the patient's home for efficient service delivery. Workload balancing strategies are incorporated, taking into account factors like total service and idle times, restricted daily working hours, flexible start times, and the minimization of idle costs, with the aim of enhancing caregiver satisfaction.

3.2. Model formulation

In this subsection, the stochastic mathematical model for the proposed home healthcare routing and scheduling problem regarding the aforementioned challenges is presented. This model is defined on a

complete connected graph $G = (M, E)$, where M includes all nodes such as nodes related to the company center (a, e) that both of them are in the same location, patients' houses (La, Nc), laboratory (La), nurses' houses (Lk, Rk) and the lunch breaks of each nurse (LB). $E = \{(i, j) | i, j \in M, i \neq j\}$ is the set of edges. Nr and Nc represent the location of non-corona and corona patients' homes, respectively. This problem is described as daily planning. A set of caregivers K is available to provide services for patients. These caregivers are divided into two groups, Kc related to caregivers who are in their corona shift and patients who are in the Nc set have to be assigned them. Kr related to caregivers who are in their non-corona shift, and the patients who are in the Nr set should be assigned to the same group. The assumptions of the proposed stochastic home healthcare optimization model are as follows:

- The planning phase incorporates known information such as the number of patients' homes, desired services, the count of nurses, their respective skills, and locations.
- The introduction of randomness is evident in the consideration of the number of demands at patients' homes, which follows a discrete random parameter.
- The service time required in patients' homes adheres to a normal distribution with the parameter $N(\mu_i, \sigma_i^2)$.
- Vehicle type three is dedicated to providing housekeeping services, encompassing tasks like food preparation and laundry services.
- All patient requests related to Home Delivery Services (HDSs), including medicine preparation, food provisioning, and laundry services, are prearranged at the company's center and delivered to patients at scheduled times.
- Predetermined time windows for each patient must be strictly adhered to.

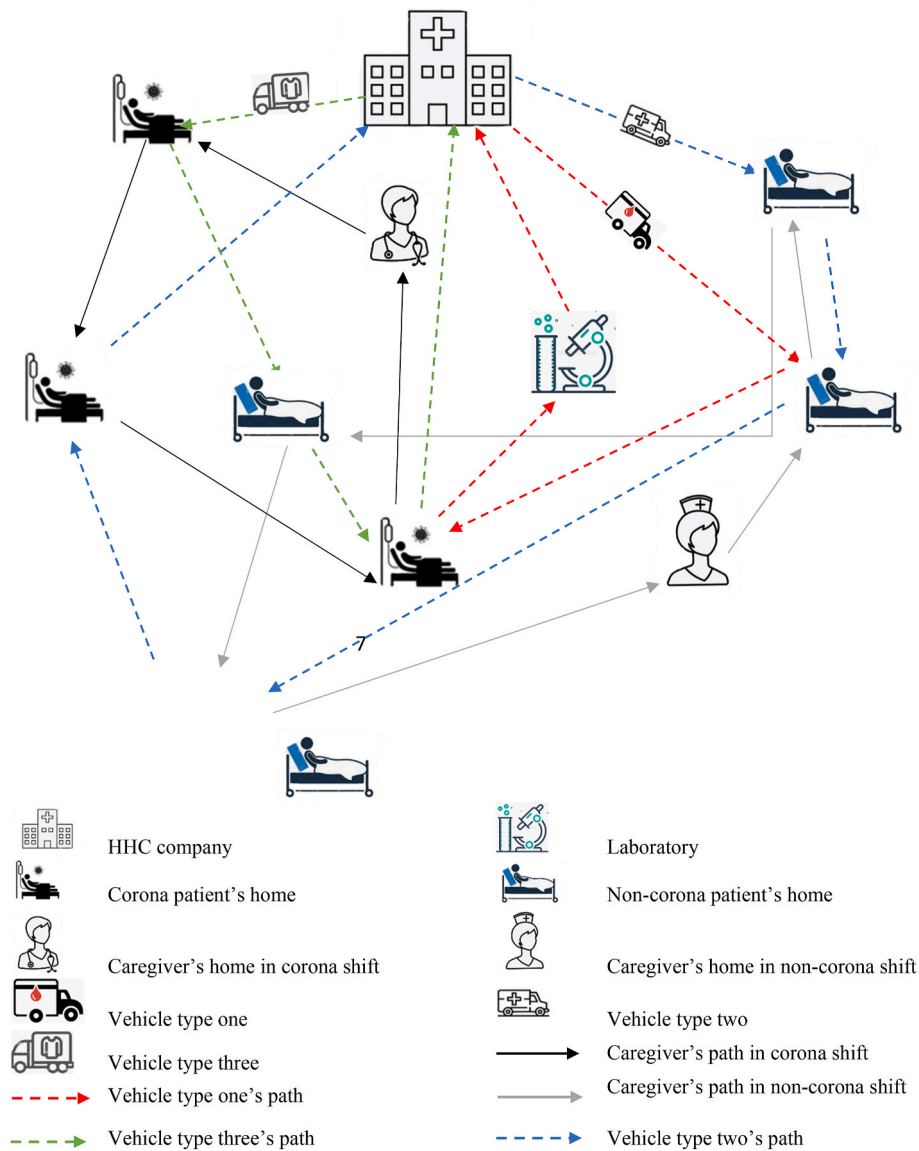


Fig. 1. An illustrative example of the proposed structure for the on-hand problem.

- The anticipated traveling times between nodes are known in advance.
- The heterogeneity of vehicles is a key consideration, encompassing variations in service time and duties.
- Each nurse is assigned her own vehicle, and concepts such as travel sharing, and multi-mode traveling are not part of the consideration.
- The study incorporates the random demand initiated by patients, aligning it with the skill set of the assigned nurses and vehicles.

The formulation of the stochastic mathematical model commences with the definition of indices and decision variables for both the first and second stages. Subsequently, the stochastic model is delineated, wherein the decision-making process unfolds in two stages. Initially, the decision maker allocates nurses to patients and determines the optimal routing for nurses. Following the identification of various scenarios, the subsequent step involves decision-making on appointment scheduling. This structured approach aims to address the dynamic and uncertain aspects of the planning process.

Notations

3.2.1. Indices

i, j, r	Index of nodes.
k	Index of caregivers.
v	Index of vehicles.
ξ	Index of scenarios related to stochastic service time.
λ	Index of scenarios related to the number of demands.

3.2.2. Sets

M	Set of all nodes, $M = \{\theta, e, Nr, Nc, La, Lk, Rk, LB\}$.
K	Set of caregivers.
Kr	Set of caregivers who are in the shift serving non-corona patients.
Kc	Set of caregivers who are in the shift serving Corona patients.
N	Set of all patients.
Nr	Set of non-corona patients.
Nc	Set of Corona patients.
S	Set of care services provided by company.

(continued on next page)

(continued)

L_k	Set of caregivers' home addresses (L_k start node for caregiver k , R_k end node for caregiver k).
LB	Set of lunch breaks for each caregiver.
V	Set of all vehicles ($Ve1 \cup Ve2 \cup Ve3$).
$Ve1$	Set of vehicles type one.
$Ve2$	Set of vehicles type two.
$Ve3$	Set of vehicles type three.
Vs	Set of services provided by the vehicles.
$\mathcal{B}$	Set of scenarios deal with stochastic service time.
ψ	Set of scenarios related to the number of stochastic demands.

3.2.3. Parameters

t_{ij}	Travel time between node i and j
td_{is}	Required time to provide care service s to patient i by each caregiver.
tdv_{is}	Required time to provide vehicles' services to patient i .
d_{ij}	Distance between node i and j
$[l_i, u_i]$	The predetermined time window of patient i , l_i is the lower bound on the patient time window, u_i is the upper bound on the patient time window.
a_{ks}	Input matrix of caregiver qualifications, where 1 means that caregiver k has the qualification of doing service s .
av_{vs}	Input matrix of HDV services, where 1 means that vehicle v provides service s .
dm_{is}	Input matrix of patient's service requirements, where 1 means that patient j needs service s .
wl	Maximum working time for each caregiver during a working day.
$[e_k, f_k]$	The predetermined time window by each caregiver who starts and finishes his working hours in this domain. e_k is the earliest time that the caregiver k starts to work, and f_k is the latest time that caregiver k finishes.
ef_c	Effective work rate for Corona caregivers.
ef_r	Effective work rate for non-corona caregivers
cv_v	Variable cost of vehicles
cit	Cost of the caregiver's idle time
c_v	Fixed cost for vehicle v
c_{tt}	Cost of moving caregiver between nodes
mw	The total working time of caregivers during a working day
tr	The maximum length of the route that each vehicle is allowed to travel during the day.
$prob_{\xi}$	The probability of the scenarios related to stochastic service time occurring.
$\eta_{i\xi}$	Coefficient of service time to the patient i under scenario ξ .
$probd_{\lambda}$	The probability of each scenario deal with the number of demands.

3.3. Decision variables

3.3.1. First stage decision variables

x_{ijk}	1, if nurse $k \in Nr$ moves along arc (i, j)
y_{ijk}	1, if nurse $k \in Nc$ moves along arc (i, j)
z_{ijv}	1, if vehicle $v \in V$ moves along arc (i, j)
y_{ev}	1, if the vehicle v leaves the company and visits patients

3.3.2. Second stage decision variables

$at_{ik\xi\lambda}$	The start time of nurse $k \in K$ at node $i \in N$ under scenario $\xi \in \mathcal{B}$ and $\lambda \in \psi$.
$tv_{iv\xi\lambda}$	The start time of vehicle $v \in V$ at node $i \in N$ under scenario $\xi \in \mathcal{B}$ and $\lambda \in \psi$.
$idt_{k\xi\lambda}$	The duration of the nurse $k \in K$ being unemployed under scenario $\xi \in \mathcal{B}$ and $\lambda \in \psi$.
$tst_{k\xi\lambda}$	The total service time of the nurse $k \in K$ during one day under scenario $\xi \in \mathcal{B}$ and $\lambda \in \psi$.

3.4. The mathematical model

3.4.1. Objective function

$$f : \text{Min } w = \sum_{v \in V} \sum_{i \in M} \sum_{j \in M} cv_v * d_{ij} * z_{ijv} + \sum_{v \in V} c_v * y_e + \sum_{i \in N} \sum_{j \in N} \sum_{k \in K} (x_{ijk} * t_{ij} + y_{ijk} * t_{ij}) * c_{tt} + \sum_{k \in K} \sum_{\xi \in \mathcal{B}} \sum_{\lambda \in \psi} prob_{\xi} * probd_{\lambda} * idt_{k\xi\lambda} * cit \quad (1)$$

The objective function is stated in Equation (1) and minimizes the

variable and fixed costs of the vehicles as well as the total cost of the nurses' traveling cost, and the nurses' unemployment time during the day by considering different scenarios.

$$\sum_{j \in Nr} x_{ijk} = \sum_{j \in Nr} x_{jik} = 1 \quad \forall k \in Kr, i \in L_k, R_k \text{ respectively} \quad (2)$$

$$\sum_{j \in Nc} y_{ijk} = \sum_{j \in Nc} y_{jik} = 1 \quad \forall k \in Kc, i \in L_k, R_k \text{ respectively} \quad (3)$$

$$\sum_{j \in Nr} x_{jik} = \sum_{j \in Nc} y_{jik} = 0 \quad \forall i \in L_k, k \in Kr, Kc \text{ respectively} \quad (4)$$

$$\sum_{j \in Nr} x_{ijk} = \sum_{j \in Nc} y_{ijk} = 0 \quad \forall i \in R_k, k \in Kr, Kc \text{ respectively} \quad (5)$$

Constraints (2), (3), (4), and (5) are used to guarantee that the starting and ending place of each nurse is their own home.

$$\sum_{i \in N} z_{aiv} = \sum_{i \in N} z_{iev} = 1 \quad \forall v \in Ve1 \quad (6)$$

Constraint (6) ensures that each vehicle starts and ends in its direction from the company.

$$\sum_{i \in N} z_{ilav} = 1 \quad \forall v \in Ve1 \quad (7)$$

$$z_{laev} = 1 \quad \forall v \in Ve1 \quad (8)$$

Constraints (7) and (8) state that the first type of vehicle before ending its route at the company center, first goes to the designated laboratory to deliver the laboratory samples and returns to the company center after delivering the samples.

$$\sum_{i \in Nr, k \in Kr} x_{ijk} = \sum_{i \in Nc, k \in Kc} y_{ijk} = 1 \quad \forall j \in Nr, Nc \text{ respectively}, i \neq j \quad (9)$$

Constraint (9) guarantees that only one nurse is assigned to each patient.

$$\sum_{i \in Nr} x_{ijk} - \sum_{i \in Nr} x_{jik} = 0 \quad \forall j \in Nr, k \in Kr \text{ and } i \neq j \quad (10)$$

$$\sum_{i \in Nc} y_{ijk} - \sum_{i \in Nc} y_{jik} = 0 \quad \forall j \in Nc, k \in Kc \text{ and } i \neq j \quad (11)$$

$$\sum_{i \in N} z_{ijk} - \sum_{i \in N} z_{jik} = 0 \quad \forall j \in N, v \in V \text{ and } i \neq j \quad (12)$$

Constraints (10), (11), and (12) indicate the continuation of the route so that when a nurse or a vehicle enters the patient's home to provide services, it must leave that place after completing the service and go to the next patient's house.

$$\sum_{i \in Nr} x_{ijk} * dm_{js} \leq a_{ks} \quad \forall j \in Nr, k \in Kr, s \in S \quad (13)$$

$$\sum_{i \in Nc} y_{ijk} * dm_{js} \leq a_{ks} \quad \forall j \in Nc, k \in Kc, s \in S \quad (14)$$

Inequalities (13) and (14) state that when a patient requests services, a nurse must be assigned to this patient who has the necessary skills to perform the desired service.

$$\sum_{i \in N} z_{ijv} * dm_{js} \leq av_{vs} \quad \forall j \in N, s \in S, v \in V \quad (15)$$

Inequality (15) indicates that a vehicle sent to fulfill a patient's request must have the ability to perform that service.

$$\sum_{i \in Nr} x_{ijk} = \sum_{i \in Nr} x_{jik} = 1 \quad \forall k \in Kr, j \in LB \quad (16)$$

$$\sum_{i \in Nc} y_{ijk} = \sum_{j \in Nc} y_{jik} = 1 \quad \forall k \in Kc, j \in LB \quad (17)$$

Constraints (16) and (17) show that a lunch break time, considered as a dummy node, is assigned to each nurse.

$$at_{ik\xi\lambda} + 180 \leq at_{jk\xi\lambda} \leq at_{rk\xi\lambda} - 180 \quad \forall i \in L_k, r \in R_k, k \in K, j \in LB, \xi \in \mathbb{B}, \lambda \in \psi \quad (18)$$

Constraint (18) specifies the lunch time of nurses under different scenarios. It indicates that the lunch break should commence no earlier than 3 h after the nurses begin their shift and no later than 3 h before the conclusion of their work, aligning with the time when the nurses finish their day and arrive home.

$$at_{ik\xi\lambda} + \eta_{i\xi} td_{is\lambda} + t_{ij} - at_{jk\xi\lambda} \leq (1 - x_{ijk}) * Q \quad \forall i, j \in Nr, k \in Kr, \xi \in \mathbb{B}, \lambda \in \psi, s \in S, i \neq j, Q \text{ is a big number} \quad (19)$$

$$at_{ik\xi\lambda} + \eta_{i\xi} td_{is\lambda} + t_{ij} - at_{jk\xi\lambda} \leq (1 - y_{ijk}) * Q \quad \forall i, j \in Nc, k \in Kc, \xi \in \mathbb{B}, \lambda \in \psi, s \in S, i \neq j, Q \text{ is a big number} \quad (20)$$

Constraints (19) and (20), indicate that the start time of the new service should be after the end of the previous service in addition to the required time for transferring the nurse to the new place.

$$l_i \leq at_{ik\xi\lambda} \leq u_i \quad \forall i \in N, k \in K, \xi \in \mathbb{B}, \lambda \in \psi \quad (21)$$

Constraint (21) shows that the time to start serving the patient should be in the minimum and maximum time period coordinated with the patient.

$$tv_{iv}\xi\lambda + td_{vis} + t_{ij} - tv_{jv}\xi\lambda \leq (1 - z_{ijv}) * Q \quad \forall i, j \in N, v \in V, \xi \in \mathbb{B}, \lambda \in \psi, i \neq j, Q \text{ is a big number} \quad (22)$$

Constraints (22), indicates that the start time of the new service provided by each vehicle should be after the end of the previous service in addition to the required time for transferring the vehicle to the new place.

$$at_{ik\xi\lambda} \leq tv_{iv}\xi\lambda \leq at_{ik\xi\lambda} + \eta_{i\xi} td_{is\lambda} \quad \forall i \in N, k \in K, s \in S, \xi \in \mathbb{B}, \lambda \in \psi \quad (23)$$

Constraint (23) refers to the simultaneous presence of the nurse and the vehicle at the patient's home under scenario ξ . Here, for nurses to be able to use medical equipment to perform their services, and for nurses to fully supervise the services provided by the vehicles, the required vehicles must arrive at the patient's home when the nurse is at the patient's home.

$$\sum_{j \in N} z_{jv} \leq y_{ev} \quad \forall v \in V, j \in N \quad (24)$$

Constraint (24) shows whether the vehicle leaves the company to serve the patients.

$$idt_{k\xi\lambda} = mwt - \left(\sum_{i \in Nr} \sum_{\xi \in \mathbb{B}} \sum_{\lambda \in \psi} td_{is\lambda} * \eta_{i\xi} * x_{ijk} + \sum_{i \in Nr} t_{ij} * x_{ijk} \right) \quad \forall i, j \in Nr, k \in Kr, \xi \in \mathbb{B}, \lambda \in \psi, s \in S \quad (25)$$

$$idt_{k\xi\lambda} = mwt - \left(\sum_{i \in Nc} \sum_{\xi \in \mathbb{B}} \sum_{\lambda \in \psi} td_{is\lambda} * \eta_{i\xi} * y_{ijk} + \sum_{i \in Nc} t_{ij} * y_{ijk} \right) \quad \forall i, j \in Nc, k \in Kc, \xi \in \mathbb{B}, \lambda \in \psi, s \in S \quad (26)$$

Constraints (25) and (26) determine the time of the nurse's unemployment under different scenarios.

$$at_{jk\xi\lambda} - at_{ik\xi\lambda} \leq wlv \quad \forall i \in L_k, j \in R_k, \xi \in \mathbb{B}, \lambda \in \psi, k \in K \quad (27)$$

Constraint (27) indicates the maximum working hours for nurses, under each scenario, which cannot exceed a certain amount according to work laws.

$$e_k \leq at_{ik\xi\lambda} \leq at_{jk\xi\lambda} \leq ef_k \quad \forall k \in K, i \in L_k, j \in R_k, \xi \in \mathbb{B}, \lambda \in \psi, k \in K \quad (28)$$

Constraint (28) shows that each nurse determines the time window that he/she can provide to the company to serve patients; thus, the patients must be assigned to the nurse in this time window under each scenario.

$$\sum_{i \in Nr} x_{ijk} * td_{is\lambda} * \eta_{i\xi} \geq ef_r * mwt \quad \forall k \in Kr, j \in Nr, s \in S, \xi \in \mathbb{B}, \lambda \in \psi \quad (29)$$

$$\sum_{i \in Nc} y_{ijk} * td_{is\lambda} * \eta_{i\xi} \geq mwt * ef_c \quad \forall k \in Kc, j \in Nc, s \in S, \xi \in \mathbb{B}, \lambda \in \psi \quad (30)$$

Constraints (29) and (30) show the workload balancing in such a way that the ef_c and ef_r factors are calculated according to the total volume of workload on a given day and the number of nurses in the company to serve patients, and the total service time of the nurses, under each scenario, must not be less than a specific percent of the total working time.

$$\sum_{i \in M} z_{ijv} * d_{ij} \leq tr \quad \forall v \in V, j \in M \quad (31)$$

Constraint (31) shows that the length of each vehicle's route during the day should not exceed a certain amount.

$$at_{ik\xi\lambda}, tv_{iv}, \xi\lambda idtk\xi\lambda \geq 0 \quad (32)$$

$$x_{ijk}, y_{ijk}, z_{ijk}, y_{ev} \in \{0, 1\} \quad (33)$$

Constraints (32) and (33) ensure the feasibility of decision variables for the proposed model.

4. Solution approach

4.1. An overview of stochastic programming

The inception of stochastic planning, introduced by Dantzig in 1970, provides a systematic framework for addressing uncertainties in decision-making. Within this approach, uncertainty is conceptualized as distinct scenarios, each characterized by a known probability of occurrence. Despite its conceptual appeal, this approach grapples with challenges, notably the intricate task of precisely determining data distributions and the exponential growth of problem dimensions with an escalating number of scenarios [55]. Decisions within stochastic programming problems are categorized into two stages: the first stage involves decisions made prior to conducting experiments, while the second stage encompasses decisions made subsequent to the experiment, reflecting a dynamic and adaptive decision-making process. The decisions of the first stage are usually represented by the vector x and the decisions of the second stage by the vector y or $y(\omega)$ or $y(\omega, x)$. In summary, the stochastic planning decisions can be displayed in the form of relation:

$$X \rightarrow \xi(\omega) \rightarrow Y(\omega, x) \quad (34)$$

Stochastic programming problems are generally defined as relation

(35):

$$\begin{aligned} & \text{Min } C^T X + E_{\xi} Q(x, \xi) \\ & \text{s. t. } Ax = b \\ & \quad x \geq 0 \end{aligned} \quad (35)$$

where $Q(x, \xi) = \min \{q^T y \mid Wy = h^T - Tx, y \geq 0\}$ and the vector ξ consists of components q^T , h^T and T . Also, E_{ξ} represents the mathematical expectation with respect to ξ . For more study in the field of stochastic programming, that can be referred to the book Introduction to stochastic programming written by Birge and Louveaux [56].

4.2. An overview of K-means clustering

The K-Means clustering algorithm stands out as a prominent and straightforward clustering technique with a history spanning over 50 years. Recognized for its ease of implementation, simplicity, efficiency, and empirical success, K-Means remains a widely applied clustering algorithm across diverse scientific domains. The fundamental steps of its implementation [57] are outlined as follows.

- First step: The initial dataset is partitioned into K initial clusters.
- Second step: A new partition is created by assigning each data point to the cluster with the closest center.
- Third step: The centers of the new clusters are calculated.
- Fourth step: The process iterates until stability is achieved in the clusters during the second and third steps.

This research employs the K-Means clustering method as a strategic approach to alleviate the challenges associated with an overwhelming number of scenarios in stochastic planning problems. Decision makers frequently encounter a vast array of potential scenarios, making it impractical to address stochastic planning comprehensively within such a large scenario pool. Consequently, this study focuses on the reduction of generated scenarios. A key feature of this approach lies in the condensed nature of the resulting scenarios, capturing a synthesis of information from all scenarios. This outcome is achieved through the iterative recalculation of cluster centers at each step of the algorithm.

4.3. Expected value of perfect information

The Expected Value of Perfect Information (EVPI) serves as a metric to quantify the highest value a decision maker is willing to invest in acquiring accurate and comprehensive information about future uncertainties. Originating from the realm of decision analysis, the concept of EVPI was initially introduced by Raiffa and Schlaifer in 1961 [58]. In the context of stochastic planning, uncertainty is conceptualized as an array of scenarios, with the random variable ξ employed to denote each distinct scenario. The formulation of the random programming model, denoted as model (36), incorporates this variable to represent the various scenarios under consideration:

$$\min z(x, \xi) = c^T x + \min \{q^T y \mid Wy = h - Tx, y \geq 0\} \quad (36)$$

$$\text{s.t. } Ax = b, x$$

By considering $\bar{x}(\xi)$ as the optimal solution derived from the random model, and $z(\bar{x}(\xi), \xi)$ as the solution obtained from the objective function, the combined solution of expectation and observation, referred to as WS (Wait-and-See), can be formulated as expressed in equation (37). The Wait-and-See response, or Here-and-Now, entails deriving an answer through mathematical expectation based on the responses generated across different scenarios. The expected value is computed in accordance with the definition, quantifying the disparity between the solution of the recourse problem (RP) and the outcome of Wait-and-See (WS) (38). Additionally, the solution to the recursive problem is determined according to Equation (39):

$$WS = E_{\xi}[\min z(x, \xi)] = E_{\xi}z(\bar{x}(\xi), \xi) \quad (37)$$

$$EVPI = RP - WS \quad (38)$$

$$RP = \min E_{\xi}z(x, \xi) \quad (39)$$

4.4. Value of stochastic solution

In practice, computing the WS response is often unattainable due to the absence of complete information. To simplify this challenge, a proposed approach involves addressing a more manageable problem by considering the mathematical expectation of each random variable instead of their actual values, yielding the Expected Value (EV) as denoted in Equation (40). Here, $\bar{\xi}$ signifies the mathematical expectation of the random parameters. By evaluating the EV across various scenarios, the response to EEV (Expectation of the Expected Value) is determined according to equation (41). Subsequently, the stochastic response value is derived as the discrepancy between the solution to the Recourse Problem (RP) and EEV, as illustrated in Equation (42).

$$EV = \min(x, \bar{\xi}) \quad (40)$$

$$EV = E_{\xi}(z(\bar{x}(\bar{\xi}), \bar{\xi})) \quad (41)$$

$$VSS = EEV - RP \quad (42)$$

5. Computational results

In this section, the efficacy and accuracy of the proposed model are scrutinized through the resolution of a small-scale example. Notably, there is a dearth of analogous models in the existing literature, prompting the utilization of Nasir et al., 's 2020 [9] example to validate the proposed model. Since certain requisite information for solving the model is absent in the referenced example, random data generation is employed to supplement the necessary details. All computations in this section are executed using the IBM ILOG CPLEX Optimization Studio version 12.6.0.0 software, running on a computer equipped with an Intel (R) Core (TM) i7-10750H CPU @ 2.60 GHz 2.59 GHz and 8 GB RAM.

5.1. Example specification

In this illustrative case, the scenario involves three nurses, with one designated for corona-related duties, catering specifically to corona patients, while the others are assigned to non-corona shifts, attending to patients without such health concerns. Each nurse possesses distinct skills, detailed in Table 2. As previously elucidated, the home health care company features three diverse vehicles, each dedicated to specific services, as outlined in Table 3.

The diverse categories of care services offered by nurses to patients are outlined in Table 4. Concurrently, Table 5 delineates the various types of vehicle services, which encompass both distinct services like food and medicine delivery, laundry services, and supplementary services that are coordinated with the care provided by nurses, necessitating their simultaneous presence at patients' homes.

Each patient needs a set of specific services and time window, which are listed in Table 6.

The traveling times between two nodes for non-corona and corona patients are given in Tables 7 and 8, respectively.

Table 2
Skills of nurses.

Nurses	Skills
Kc1	S3, S4, S5, S6
Kr2	S1, S3, S4, S5, S6
Kr3	S2, S3, S5

Table 3
Services of vehicles.

Vehicles	Services
V1	Vs1
V2	Vs2, Vs3
V3	Vs4, Vs5

Table 4
Type of care services.

Service title	Service description	Duration of service
S1	Speech therapy	60
S2	Dressing and injections	30
S3	Insulin injection	30
S4	Laboratory sampling	30
S5	physiotherapy	90
S6	X-ray imaging	30

Table 5
All types of vehicle services.

Service title	Service description	Duration of service
Vs1	Delivery of laboratory samples	15
Vs2	Delivery of medicine to patients	15
Vs3	Providing medical equipment needed by nurses	30
Vs4	Meal delivery	15
Vs5	Laundry services	15

Table 6
Properties of patients in the considered example.

	Patients	Required services	Acceptable time window
Non-corona patients	Pa1	S4, Vs1	[345, 465]
	Pa3	S2, Vs4	[30, 150]
	Pa5	S3, Vs5	[254, 375]
	Pa6	S5, Vs4, Vs5	[184, 304]
	Pa9	S1, S4, Vs1	[298, 418]
Corona patients	Pa10	S3, S6, Vs2	[148, 268]
	Pa2	S5, Vs2	[185, 305]
	Pa4	S4, Vs1	[60, 180]
	Pa7	S3, Vs4	[434, 554]
	Pa8	S5, S6, Vs3	[46, 166]

In order to generate a scenario for the random parameters of the problem, which is the number of requests and also the time required to provide the service to the patients, is done.

5.2. Introducing scenarios

5.2.1. Scenarios for demand numbers

Given the discrete nature of the number of requests, informed by experiences during the Corona era and the inherent randomness in the

Table 7
Traveling times between two nodes for non-corona patients.

	Kr2	Kr3	Pa1	Pa3	Pa5	Pa6	Pa9	Pa10	La	ə
Kr2	0	0	19.2	20.8	8.5	18	17.5	21.7	16.3	0
Kr3		0	22.6	11.7	22.5	27.8	26.1	29.6	24.2	0
Pa1			0	19.1	17.4	19.7	25.5	20.8	11.7	16.5
Pa3				0	19.5	25.1	24.9	26.8	19.1	10.9
Pa5					0	17.5	19.3	21.1	12.2	19.4
Pa6						0	21.1	22.7	22.5	24.2
Pa9							0	25.7	25.9	25.9
Pa10								0		25.4
La									0	10.3
ə										0

requests at a patient's home, particularly within a megacity like Tehran, a discrete probability distribution is employed. The scenarios associated with varying demand levels are outlined in Table 9.

5.2.2. Scenarios for service time coefficients

Conversely, as time is a continuous parameter following a normal distribution $\tilde{N}(1,0.2)$, a substantial number of scenarios arises, making direct model resolution impractical. To address this, an initial set of 1000 scenarios is randomly generated, and then the k-means clustering algorithm is applied to condense this set, aiming to retain essential information while reducing the computational burden. To determine the optimal number of scenarios, sensitivity analysis is conducted. Subsequent to this analysis, the scenarios are streamlined to the determined optimal number using the k-means algorithm. All computations in this section are executed using Python software on a computer equipped with an Intel(R) Core(TM) i7-10750H CPU @ 2.60 GHz processor and 8 GB of RAM.

5.2.3. Sensitivity analysis on the number of optimal scenarios

To determine the optimal number of scenarios for solving the problem, a sensitivity analysis is conducted on the K parameter of the K-means algorithm. The focus of this analysis is on service time, given its continuous nature and common representation as a normal distribution with a mean of 1 and a standard deviation of 0.2 in literature.

Initially, 1000 scenarios are generated based on this distribution, and the K-means clustering algorithm is applied to cluster these scenarios into k clusters using Python. The process involves determining the number of centroids and subsequently assigning the number of members in each cluster with known centroids. To ascertain the optimal number of clusters (K), the model is solved for different numbers of clusters (ranging from 1 to 10), with the results plotted in Fig. 2. The analysis reveals that as the number of clusters increases, both the value of the

Table 8
Traveling times between two nodes for corona patients.

	Kc1	Pa2	Pa4	Pa7	Pa8	La	ə
Kc1	0	11.1	24.5	24.3	18.8	15.3	0
Pa2		0	23.6	26	15.4	22.1	23.5
Pa4			0	14.9	22.5	19.5	16.7
Pa7				0	28	13.5	19.2
Pa8					0	17.6	24.2
La						0	10.3
ə							0

Table 9
Random demand number scenarios.

	Number of scenarios		
	Se1	Se2	Se3
No. of requests	1	2	3
Probability	0.5	0.3	0.2

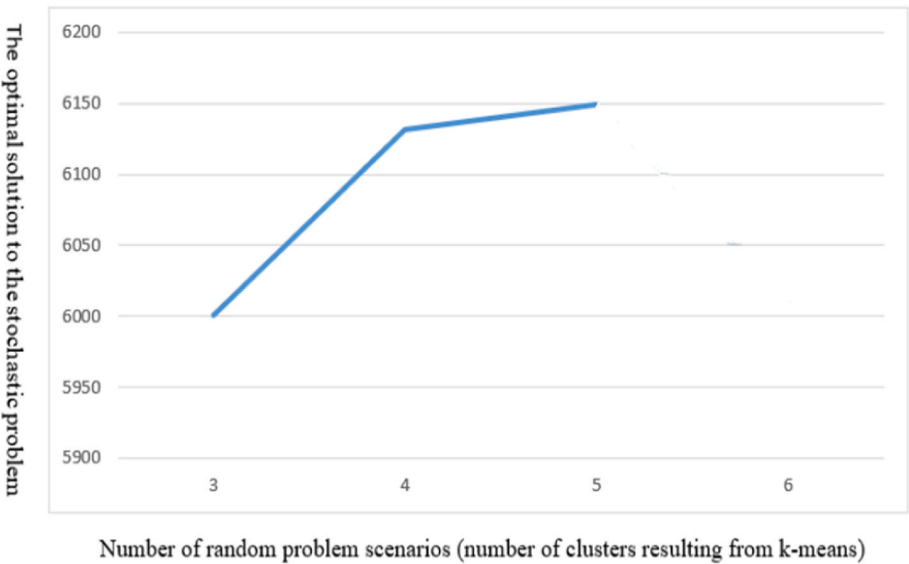


Fig. 2. Sensitivity analysis on the number of optimal scenarios.

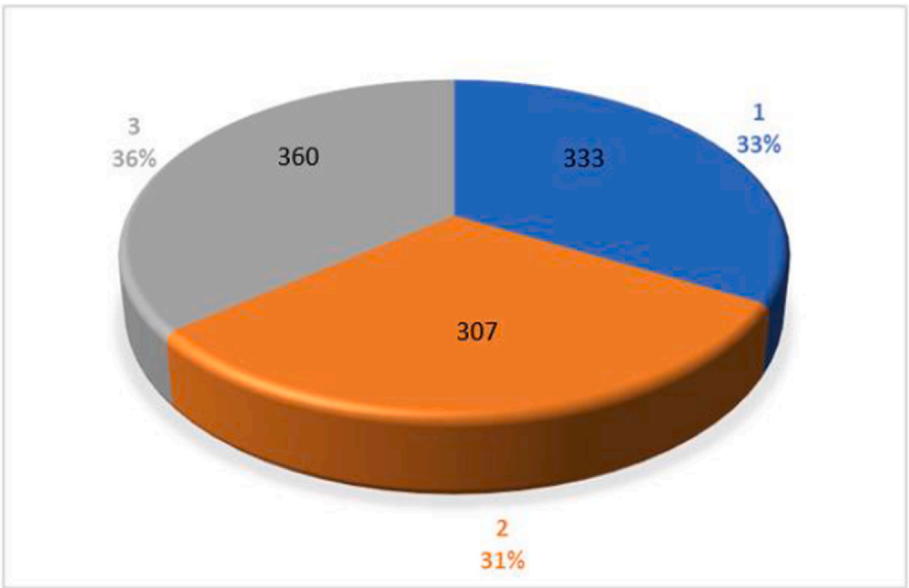


Fig. 3. The total number of scenarios in each cluster and the probability of each cluster.

Table 10
Specification of reduced scenarios with K-Means algorithm (k = 3).

Scenario No.	Probability	Service coefficients for patients									
		Pa1	Pa2	Pa3	Pa4	Pa5	Pa6	Pa7	Pa8	Pa9	Pa10
1	0.33	1	1.1	1.02	0.99	1.02	0.8	1.11	0.8	0.98	0.8
2	0.31	0.95	1	0.8	1.04	0.8	1.1	0.8	1.02	0.8	1.1
3	0.36	1.1	0.8	1.1	0.95	1.1	1	0.99	1	1.02	1

objective function and the time required for problem-solving also increase. Consequently, K is chosen to be 3 for further analysis. Applying the K-Means algorithm with K = 3 on the existing 1000 scenarios, Fig. 3 illustrates the distribution of scenarios in each cluster along with the probability of occurrence for each cluster. Detailed results for each cluster are presented in Table 10.

5.3. Results of stochastic programming model

In this section, the efficacy of the newly introduced stochastic model is demonstrated through the utilization of five distinct examples from the computational dataset provided in Nasir et al.’s publication [9]. These examples involve the consideration of 10 patients and 3 nurses, with a classification of patients and nurses into two groups—corona and non-corona—explored. Given the nature of the disease and the unique

Table 11
Results of the stochastic programming model.

Sample No.	Optimal answer	Computational times (sec.)
1	6055.99	1.53
2	5758.48	3.4
3	5963.4	2.6
4	5781.3	2.8
5	6043.5	1.82

challenges posed by the COVID-19 outbreak, the number of demands in patients' homes, and consequently, the service times for providing services, are treated as random variables.

Additional specifications of the examples are detailed in Section 5.1. Random scenario generation and the application of K-Means Clustering are employed to establish the coefficients for service times and the probability distribution for each cluster. All computations in this section are performed using IBM ILOG CPLEX Optimization Studio version 12.6.0.0 software on a computer with an Intel(R) Core (TM) i7-10750H CPU @ 2.60 GHz, 2.59 GHz, and 8 GB RAM. The outcomes of solving the stochastic model are presented in Table 11, demonstrating that the small example of the stochastic model can be efficiently solved by the software within a reasonable timeframe.

5.3.1. Comparison of the results in the presence and absence of uncertainty

In this segment, we undertake a comparative analysis between the outcomes obtained under conditions accounting for uncertainty and those derived without such consideration. To facilitate this comparison, we utilize the data from the example expounded in Section 5.1. The optimal routes for each nurse and vehicle, as obtained from the model without incorporating uncertainty, are succinctly presented in Table 12, with the corresponding optimal value of the objective function amounting to 5283.3. The optimal route obtained from the model considering the uncertainty is given in Table 13, with the objective function of 6055.99.

As delineated in Tables 12 and 13, notable disparities emerge between the outcomes of deterministic and stochastic models. It can be inferred that the allocation of patients to each nurse, and consequently the optimal routes assigned to each nurse, varies significantly between these two scenarios. Additionally, the commencement times for lunch breaks for nurses 2 and 3 exhibit discrepancies in the deterministic model as compared to the stochastic counterpart. Furthermore, the introduction of randomness to certain parameters in the stochastic model results in an increase in the objective function's value. Notably, the stochastic model leads to a reduction in the number of patients assigned to each nurse due to the presence of uncertainties and additional constraints such as workload balancing and work regulations. Consequently, under these circumstances, companies may need to

employ a greater number of nurses. The results of solving the model while considering uncertain demand and service time are graphically depicted in Fig. 4 for enhanced comprehension.

As illustrated in Fig. 4, the proposed model has caregivers start and end their routes from their homes. Caregivers are assigned to specific time windows to meet with patients, and a 1-h lunch break is factored into their schedules according to the mathematical model. The vehicles, except for type one, begin and end their routes at the depot, serving each patient's requests along the way. The type one vehicles, however, must first deliver biological samples to the laboratory before returning to the depot, ensuring efficient and safe transportation of medical samples.

5.3.2. Wait-and-see solution (WS)

To compute the Wait-and-See (WS) answer, the stochastic problem is initially solved by addressing each scenario individually. Subsequently, the mathematical expectation is derived from these distinct solutions, taking into account the probabilities associated with each scenario. In this particular model, both the number of requests and the service time for each patient are treated as random variables, exerting an influence on the nurse's service time at the patient's residence. Given the consideration of three scenarios for the number of requests at the patient's home and an additional three scenarios for the service time associated with each request, a total of nine scenarios are generated. Consequently, the model is solved for these nine scenarios, yielding the following outcomes for each individual scenario. Table 14 presents the optimal value of the stochastic model for each scenario.

$$\begin{aligned}
 WS &= (0.165 \times 5262.3) + (0.155 \times 5287.1) + (0.18 \times 5262.3) \\
 &\quad + (0.099 \times 4696) + (0.093 \times 4722.4) + (0.108 \times 4657) + (0.066 \times 3777.5) \\
 &\quad + (0.062 \times 4447) + (0.072 \times 3791.5) \\
 &= 4839.9
 \end{aligned}$$

5.3.3. Expected value solution (EV)

To calculate the result of EV, we first calculate the mathematical expectation of the scenarios, which are listed in Tables 15 and 16.

According to Tables 15 and 16, the average service time according to all scenarios is written in Table 17. To calculate the Expected value (EV), we enter these values for the service time duration at each patient's home in the model and calculate its value. By solving the model using the values of Table 17, the answer of EV will be as follows.

$$EV = 4465.3$$

5.3.4. Expected value of perfect information

The expected value of perfect information (EVPI) is the difference between the answer of Recursive problem (RP) and the answer of wait-and-see (WS), which is calculated as follows:

Table 12
The optimal route without considering the uncertainty.

Nurse/vehicle	The optimal route
Nurse 1 (Kc1)	L1 → Pa4 → Pa8 → pa2 → Lb1 → Pa7 → R1
Nurse 2 (Kr2)	L2 → Pa10 → Pa6 → Pa9 → Lb2 → Pa1 → R2
Nurse 3 (Kr3)	L3 → Pa3 → Pa5 → Lb3 → R3
Vehicle 1	Depot → Pa4 → Pa9 → Pa1 → L → depot
Vehicle 2	Depot → Pa10 → Pa8 → Pa2 → depot
Vehicle 3	Depot → Pa3 → Pa6 → Pa5 → Pa7 → depot

Table 13

The optimal route with considering uncertainty.

Nurse/vehicle	The optimal route
Nurse 1 (Kc1)	L1 → Pa4 → Pa8 → Pa2 → Lb1 → Pa7 → R1
Nurse 2 (Kr2)	L2 → Pa10 → Lb2 → Pa9 → Pa1 → R2
Nurse 3 (Kr3)	L3 → Pa3 → Lb3 → Pa6 → Pa5 → R3
Vehicle 1	Depot → Pa4 → Pa9 → Pa1 → L → depot
Vehicle 2	Depot → Pa10 → Pa8 → Pa2 → depot
Vehicle 3	Depot → Pa3 → Pa6 → Pa5 → Pa7 → depot

$$EVPI = RP - WS = 6055.99 - 4839.9 = 1216.09$$

5.3.5. Value of stochastic solution

To calculate the value of stochastic solution, we should first calculate EEV. For this purpose, we put the answers obtained from EV in the stochastic model and solve the model. Here, because of some scenarios become infeasible for the values obtained from EV, thus, the model becomes infeasible. This means that the chance of a stochastic answer to solve this problem is too high, so it must be solved by stochastic programming.

6. Discussion and managerial insights

The method applied in this study is stochastic programming, which provides a systematic framework for decision-makers to make the best decisions under uncertain situations. Within this approach, uncertainty is conceptualized as distinct scenarios, each characterized by a known probability of occurrence. As distinct rational scenarios are needed in this method, K-means clustering is applied to reduce the number of scenarios and create a practical scenario set. In the sensitivity analysis context, the model is first examined with various numbers of clusters to find the optimal number of clusters (K). The analysis depicts that the value of the objective function and the time required for problem-solving increase with an increase in the number of clusters. Consequently, K is selected to be the least ($K = 3$). Then, the efficiency of the mathematical model is tested by utilizing five various examples. A comparative analysis is done between the outcomes obtained under conditions accounting for uncertainty and those derived without such consideration. The results of solving the model while considering uncertain demand and service time illustrate that uncertain parameters in the stochastic model result in an increase in the objective function's value and a decrease in the number of patients assigned to each nurse. Finally, to consider the necessity of using an uncertain model to solve the described problem, stochastic programming measures such as EVPI and VSS are computed. According to the obtained results, the chance of a stochastic answer to solve this problem is too high, and it must be solved by stochastic programming.

In Summary, this study presents an innovative HHCRSP (Home Healthcare Crew Routing and Scheduling Problem) model designed to address the challenges posed by contagious outbreaks, such as the COVID-19 pandemic, during quarantine conditions. The model introduces three key advancements: (1) it integrates quarantine and social distancing measures by designating multiple depots and categorizing nurses and patients to prevent disease transmission; (2) it incorporates stochastic parameters for demand and service times, a less common approach in existing literature; and (3) it emphasizes the synchronization of HHC (Home Healthcare) and home delivery services (HDSs), workload balancing, and work regulations, crucial for caregiver well-being during a pandemic. Additionally, the model features constraints

on lunch breaks, flexible work start times, and strict time windows, making it well-suited to address the unique challenges of crisis situations. However, a potential limitation of the study is the scope of uncertainties considered. Expanding the model to account for additional stochastic factors, such as variable travel times and the alignment between patient demand and caregiver skills, could further enhance the model's real-world applicability and effectiveness, enabling it to better reflect the dynamic and unpredictable nature of healthcare delivery during times of heightened uncertainty and disruption.

6.1. Managerial insights

In synopsis, societies throughout history have grappled with unforeseeable events like floods, earthquakes, and infectious diseases, necessitating health systems to assess conditions for enhanced response and societal health improvement. The recent global encounter with the COVID-19 pandemic has induced novel challenges for various sectors, including home healthcare companies. This study provides crucial managerial insights in response to these challenges.

- The proposed mathematical model underscores the importance of incorporating quarantine conditions and health protocols as mathematical constraints. Decision-makers and managers can strategically plan patient and nurse allocation, elevating health and satisfaction levels during the pandemic.
- Recognizing the inevitable presence of random parameters in real-world conditions, the model addresses the challenge of random demands in patients' homes during the pandemic. The consideration of random parameters, such as the number of demands and service times, enhances planning quality and mitigates the risk of exceeding coordinated patient time windows.
- Diverging from traditional routing and scheduling models, the proposed model facilitates patients in receiving both HHC and Home Delivery Services (HDSs) services. HDSs offer an avenue for patients and caregivers to avoid high-risk environments, leading to cost reductions and improved coordination under caregivers' supervision.
- Implementation of workload balancing concepts plays a pivotal role in fostering equity among caregivers, particularly during critical periods. Managers can establish effective work rate coefficients for COVID-19 and non-COVID-19 caregivers based on daily workload, mitigating costs linked to dissatisfaction and inefficiency.

By embracing these insights, home healthcare companies can navigate the complexities of the COVID-19 pandemic, enhancing operational efficiency and elevating the standard of patient care.

7. Conclusions, limitations, and future streams

Main contributions: Home Healthcare (HHC) companies play a

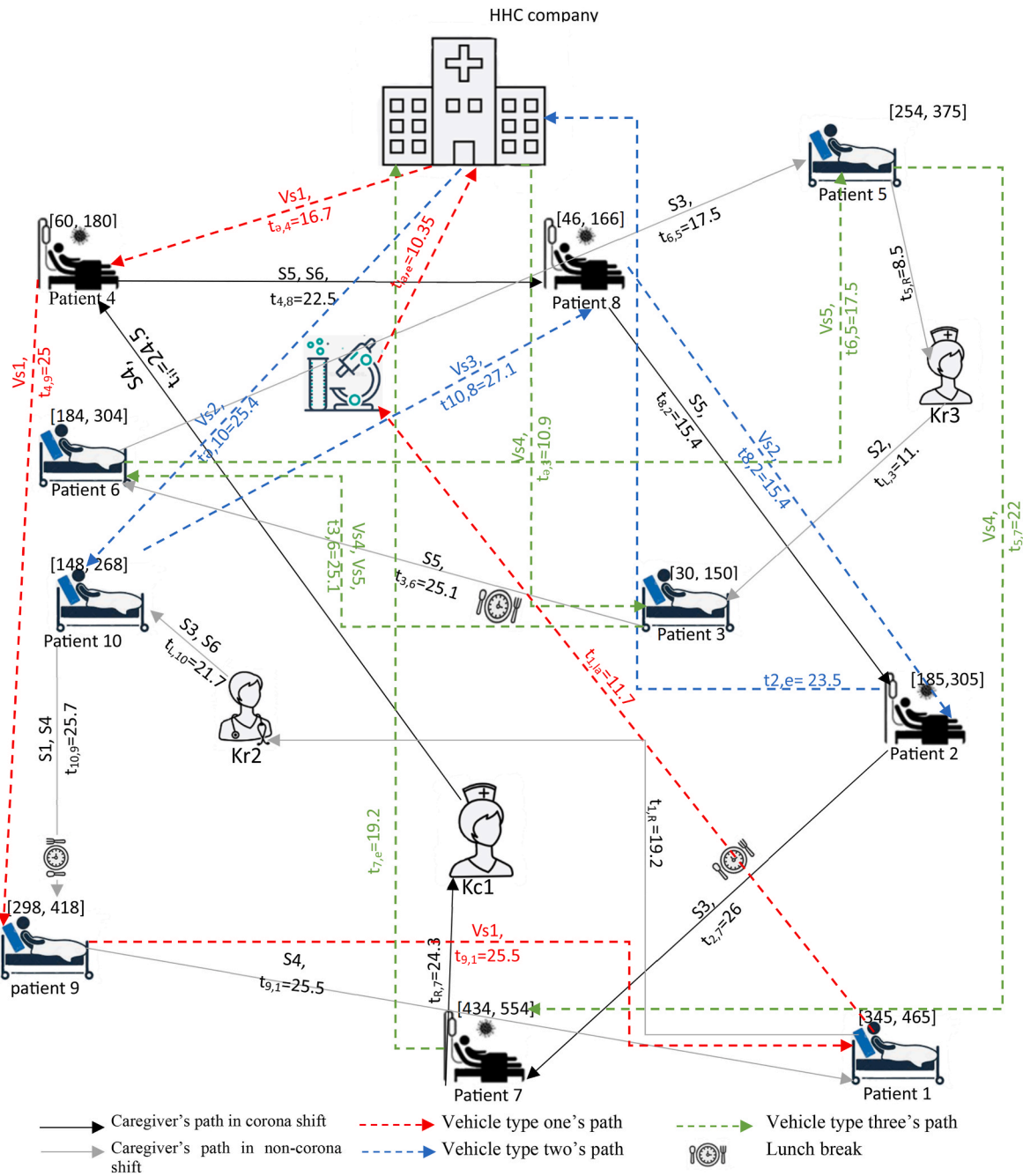


Fig. 4. The optimal route for example under uncertainty.

Table 14
The optimal value of the stochastic model for each scenario.

	S1.1	S1.2	S1.3	S2.1	S2.2	S2.3	S3.1	S3.2	S3.3
Probability	0.165	0.155	0.18	0.099	0.093	0.108	0.066	0.062	0.072
Optimal solution	5262.3	5287.1	5262.3	4696	4722.4	4657	3777.5	4447	3791.5

$S_{i,j}$: i related to demand scenarios and j related to service time scenarios.

Table 15
Service duration for average scenarios related to the number of patients' requests.

	Pa1	Pa2	Pa3	Pa4	Pa5	Pa6	Pa7	Pa8	Pa9	Pa10
Mean service duration (min)	44	90	44	44	44	81	44	81	81	90

Table 16
Average random service time coefficient for each scenario.

	Pa1	Pa2	Pa3	Pa4	Pa5	Pa6	Pa7	Pa8	Pa9	Pa10
Mean coefficients	0.9967	0.9936	0.9996	0.9972	0.9979	1.0102	0.9964	1.0063	1.002	1.0234

Table 17
The service time duration at each patient’s home.

	Pa1	Pa2	Pa3	Pa4	Pa5	Pa6	Pa7	Pa8	Pa9	Pa10
Service time duration (min)	43.9	89.4	44	43.9	43.9	81.8	43.8	81.5	81.2	92.1

vital role in delivering healthcare services directly to patients’ homes, aiming to reduce treatment costs, maintain quality care, and enhance stakeholder satisfaction. The onset of the COVID-19 pandemic prompted the World Health Organization (WHO) to issue guidelines emphasizing quarantine and social distancing measures. Adopting a "bring-service-near-your-home" policy aligns with these guidelines, facilitating service access without leaving home, especially crucial during pandemics when travel to healthcare facilities is challenging. To address pandemic challenges, a mathematical model has been proposed, optimizing personnel assignment and scheduling while considering factors such as patient classification, caregiver shifts, workload balancing, and efficient routing under uncertainties. The model incorporates WHO protocols as constraints to mitigate illness spread, with operational cost reduction as the objective function.

Results obtained from the research: This study strives to capture the realities of people’s lifestyles during the COVID-19 pandemic. Beyond adhering to quarantine conditions and workload balancing, the model addresses new challenges posed by the nature of COVID-19, such as uncertain demands in patients’ homes. This uncertainty, modeled as a known discrete distribution, coupled with uncertain service times following a continuous normal distribution, introduces complexity. Computational experiments involve two sections: the first focuses on stochastic service time, generating 1000 scenarios under a normal distribution and reducing scenarios through K-Means clustering. A sensitivity analysis on the number of clusters selects $K = 3$. Model efficiency and accuracy were then tested through small examples, considering measures like Wait and See ($WS = 4839.9$), Expected Value ($EV = 4465.3$), and Expected Value of Perfect Information ($EVPI = 1216.09$). Notably, when the Value of stochastic solution is calculated, the model becomes infeasible, indicating a high likelihood of stochastic solutions and the need for stochastic programming.

Limitations and Future research: The proposed daily mathematical programming model under uncertainties forms the basis for future research. Consideration of additional uncertainties, such as stochastic travel time and alignment of patient demand uncertainty with caregiver skills, could enhance the model’s applicability. While the proposed model is effective for small-scale problems, future research could explore heuristic and meta-heuristic algorithms for larger problems where exact methods may be impractical. The model’s extension to a multi-day planning horizon and the inclusion of additional objectives could further enhance its utility. Investigating dependencies in patients’ demands and incorporating relevant constraints would add realism to real-life applications. Meta-heuristic algorithms, including swarm-based, evolutionary-based, physics-based, game-based, and human-based methods, offer potential solutions to optimize complex problems, with ongoing research focusing on refining these algorithms for robust performance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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